

Memorandum

To: Professor Sury
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Subject: Fin 294 Problem Set 2

QUESTIONS & SOLUTIONS

4% XYZ Corp. bond due in 6 years, YTM 8% (AAA rated)

XYZ Corporation is issuing bonds to borrow money from investors. The funds raised may be used for activities such as expanding into new markets, refinancing existing debt, or covering general operating expenses. The bond is rated AAA, the highest credit rating, indicating a very low expected risk of default. Credit rating agencies such as S&P Global Ratings assign these ratings.

The 4% is the coupon rate, which determines the bond's interest payments, while the 8% YTM is the market-required annual return. The bond makes semiannual payments, meaning two payments per year, and has a par value of \$1,000, which is the amount the issuer promises to repay at maturity. The bond matures in 6 years, or 12 semiannual periods.

We use the provided assumptions of semiannual payments and a \$1,000 par value to calculate the bond's value and analyze questions a bondholder will consider when deciding whether the investment is appropriate. The results are presented through math, charts, and dashboards below to improve readability, understanding, and make the analysis easier to use.

1. What is the value of this bond?

The price of the bond is \$812.30, which we calculated using our excel model.

For each time period, we calculated the Bond Cash Flow using

```
=IFS(A14=0,0,A14<perttm,percpr*parval,A14=perttm,  
percpr*parval+parval)
```

From there we calculated the present value of the cash flow (PVCF) using

```
=B14/(1+perytm)^A14
```

where *perytm* is the Periodic Yield to Maturity, *percpr* is the Periodic Coupon Rate, *parval* is the Par Value, and *perttm* is the Periodic Term to Maturity of the bond.

We then took the sum of the PVCF for each period to give us the total price of the bond.

The price of the bond is found by taking the sum of all discounted cash flows generated by the bond, with cash flows from period 1-11 each being \$20 discounted with $(1.04)^n$, and the final cash flow at period 12 calculated by adding the par value to the coupon and discounting with $(1.04)^{12}$. Note: because this bond is compounded semiannually, we use 12 periods instead of 6.

2. Create a duration table in Excel and provide results for Macaulay duration.

The annual duration of the Bond is 5.311, as pictured below in the duration table. This value was found by summing all of the values in the $t \cdot PVF/Price$ column of the excel, which was computed using $=A14 \cdot C14 / \text{bondprice}$

Time Period	Bond Cash Flow	PV(CF)	$t \cdot PVF/Price$	DURATION Function
0	0	0	0	5.310744997
1	20	19.23076923	0.02367451	
2	20	18.49112426	0.045527903	
3	20	17.77992717	0.065665245	
4	20	17.09608382	0.084186211	
5	20	16.43854214	0.10118535	
6	20	15.80629051	0.116752327	
7	20	15.19835626	0.130972162	
8	20	14.6138041	0.143925453	
9	20	14.05173471	0.155688591	
10	20	13.51128338	0.166333964	
11	20	12.99161863	0.175930154	
12	1020	637.0889906	9.411648124	
Bond Price:		\$ 812.30	Periodic Duration:	10.621
Annual Duration:		5.311		
Annual Duration via D		5.311		

That gives you a duration of 10.62, but to annualize it, we divide by the frequency (2, since compounded semiannually). The annualized duration is 5.311 years, which represents the average time we get our money back, weighted by time and periodic cash flow.

3. Use the native Excel DURATION function to validate above results

The annual duration of the bond output from the DURATION function is 5.3107.

$=\text{DURATION}(\text{DATE}(2026,7,25), \text{DATE}(2026+\text{annttm},7,25), 0.04, 0.08, 2, 0)$

With the dates being 6 years until maturity, the annual coupon rate of 4%, annual yield to maturity being 8%, and 2 for compounding semiannually. Using this built-in function, the duration is also 5.311 years.

4. Create a data table in Excel with appropriately selected YTM's in the first column and the calculated values in the second column. Draw a bond value/yield graph with appropriate labels.

r - Annual YTM	P - Bond Price
2%	\$1,112.55
3%	\$1,054.54
4%	\$1,000.00
5%	\$948.71
6%	\$900.46
7%	\$855.05
8%	\$812.30
9%	\$772.04
10%	\$734.10
11%	\$698.35
12%	\$664.65
13%	\$632.86
14%	\$602.87

Because r , the annual YTM or required return, and P , the bond price, have an inverse relationship, we expect the graph to slope downward as r decreases. As r increases, P decreases, and as r decreases, P increases.

To test this relationship and create the graph, we first select an appropriate range of annual YTM values. We use values from 2% to 14%, which includes rates below and above the bond's current r of 8%. These values are entered in the left column of the table under r - Annual YTM.

In the right column, we calculate P reusing the formula $=PV(Perytm/freq, Perttm, Percprate*Parval, Parval)$. We slightly adjust the formula so that $Perytm$, the periodic YTM, is based on each annual YTM from the 2% to 14% range. The results confirm the expected inverse relationship: as r increases, P decreases, producing a downward-sloping curve.



Lastly, we highlight the two columns in our table and use Excel to create a scatter plot from the data. We add a line connecting the dots so the inverse relationship between r and P is easier to see. We then adjust the chart by adding an appropriate title and axis titles, removing unnecessary white space, and setting the x-axis and y-axis ranges so the data is displayed more clearly. We can also highlight important points, such as the bond price calculated in Question 1.

5. If your holding period is 4 years, what is your holding period return (HPR) on this bond if rates go up by 100 basis points (i.e., from 8% to 9%) immediately?

The total holding period return over the 4 year holding period is 35.16% which yields an annual rate of 7.82%. We got this number by taking the present value of the holding period and using that value along with the original bond price to calculate the cumulative HPR and then used that to find the annualized HPR.

$$\begin{aligned}
 &=B45/(1+pernewytm)^(A45-perholdper) \\
 &=(hprval-bondprice)/bondprice \\
 &=(1+hprcu)^(1/annholdper)-1
 \end{aligned}$$

HOLDING PERIOD RETURN CALCULATION			
	Annual	Period	
Assumed Holding Period	4	8	
Assumed Change in Yield	1%	0.005	
Old YTM	8%	0.04	
New YTM	9%	0.045	
Time Period	Bond Cash Flow	PV(CF) as of HP	
0	0	0	
1	20	27.21723661	
2	20	26.0452025	
3	20	24.92363875	
4	20	23.85037201	
5	20	22.8233225	
6	20	21.8405	
7	20	20.9	
8	20	20	
9	20	19.13875598	
10	20	18.31459902	
11	20	17.52593208	
12	1020	855.3325705	
	Value at Holding Period: \$	1,097.91	
	Cumulative HPR:	0.351611626	
	Annualized HPR:	0.078233694	

First, we calculated the value of the bond at the holding period → the sum of all present values of the cash flows $= B68 / (1 + \text{newperlytm})^{(A68 - \text{perholdper})}$, with the cell referring to the cash flow of the bond (\$20) and the period. The bond value at the holding period (year=4) is \$1,097.91. Then, to find the HPR of the bond when the change in yield is 1% (going from 8% to 9%), we used $=(\text{hprval} - \text{bondprice}) / \text{bondprice}$ where hprval is the bond value at the holding period, and bond price is the original bond price at $t=0$ (\$812.30). The HPR (cumulative) is 35.16%. We annualize it with this equation: $=(1 + C85)^{(1/\text{annholdper})} - 1 \rightarrow (1 + \text{cumulative HPR})$ converts the term into a growth factor, and using the exponent $(1/\text{annholdper})$ annualizes it by reversing compound growth. Then we subtract 1 at the end to get it back to a percentage.

6. How would you immunize yourself from the interest rate change in q5? Assume that you can sell a portion of your bond and invest in a cash (floating rate) security. What would be the percentage invested in each?

To immunize our holding-period return against the interest rate change in the previous question, we want our portfolio's weighted average duration to equal the holding period of 4 years. Right now, our duration (5.311 years) exceeds the holding period (4 years) which puts us at a loss and very sensitive to changes in interest rates. We use the formula below to calculate the exact percentage to reinvest in our corporate bond and what percentage to invest in the cash security. This means we would allocate a weight of 75.32% to the original bond while allocating the remaining 24.68% of the portfolio to a Money Market Security.

We arrived at these numbers by first identifying the two duration numbers and then using them to build an immunization formula in excel:

$$w_{\text{bond}} \cdot \text{durbond} + (1 - w_{\text{bond}}) \cdot 0 = \text{annholdper}$$

We then used this equation to solve for the weight of the original bond, and then took the complement of that value to find the weight of the money market security.

Original bond weight calculation: $=(\text{annholdper} - 0) / (\text{anndur} - 0)$

- c. **YTM (stated in APR)**
- d. **Par value (or can permit the user to simply hit enter to accept the default of \$1000)**
- e. **Compounding (or can permit the user to simply hit “enter” to accept the default semiannual compounding)**

To solve this question we used Streamlit and pandas in Python along with some HTML and CSS to build a customizable dashboard that calculates the bond price and durations based on the user's input. We also made sure to include stop checks with messages for when a user enters and incorrect input value (see photos below). The following breaks down the python files created for this dashboard:

- `bond_valuation_dashboard.py`:
 - Sets up the Streamlit dashboard and displays the user interface
 - Calls `functions.py` and `bond_valuation_calculations.py` to collect user inputs and run calculations.
 - Displays the results, plot and duration table for the given inputs.
- `bond_valuation_calculations.py`:
 - Handles all the logic for the bond calculations, verifies inputs and handles edge cases.
- `functions.py`:
 - Rendering functions to connect the calculations and the design
 - Handles user input and displays messages when wrong inputs are given
- `dashboard_design.py`:
 - Dashboard layout, buttons, colors, etc
 - Does not affect the calculations or logic of the dashboard.

https://github.com/sloholt/financial_analytics_assignments/tree/main/Assignment%202

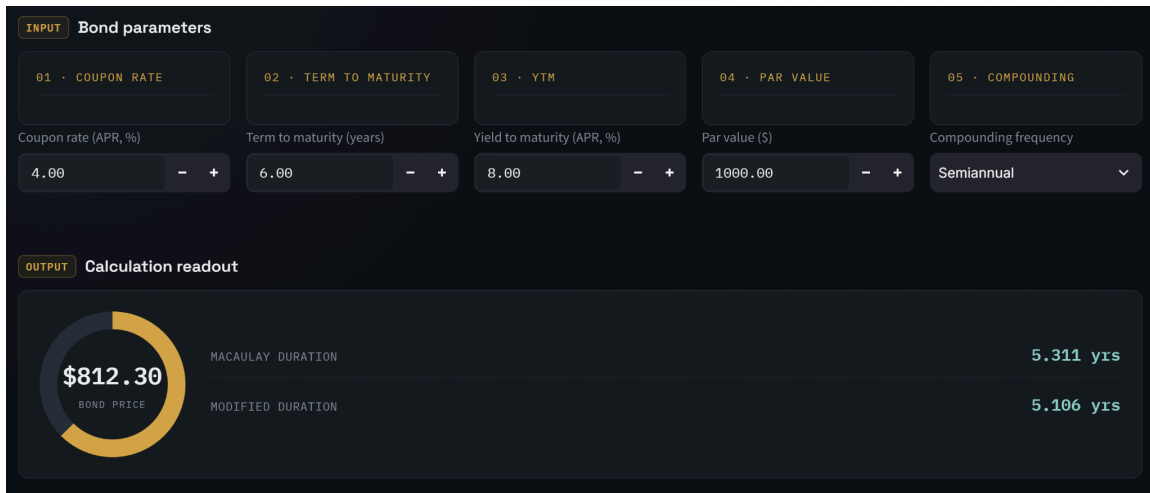
Coupon rate (APR, %) Term to maturity (years) Yield to maturity (APR, %) Par value (\$) Compounding frequency

6.00 5.10 7.00 1000.00 Semiannual

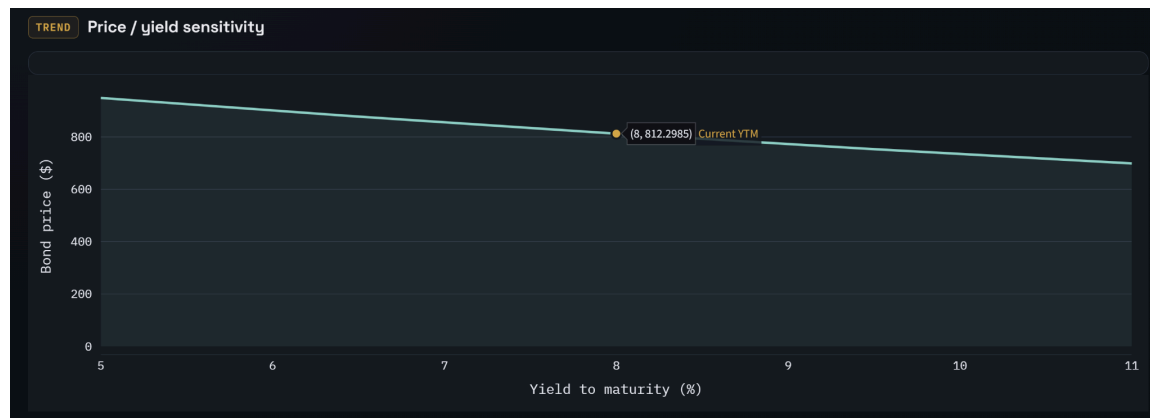
Invalid inputs: Total periods must be a whole number

9. Demonstrate that the solution is the same as your solution in Excel for the bond referenced at the beginning.

Dashboard output for the reference bond: 4% XYZ Corp. bond due in 6 years, YTM 8%



10. Draw a bond value/yield graph as part of the Python program you wrote for number 6.



LLM USAGE

Models Used: Claude, Claude Code

Prompts:

- How do I restrict and display input errors to the user given my edge cases in the calculations inside bond_value_calculations.py ?
- How do I build a duration table in python?
- How do I make a message appear when the user tries to input a number below 0 for the coupon rate, ytm and par value?
- Help me write the html for these functions to connect my calculations to my dashboard design.
- How come despite adding the clamps, it doesn't display any message when the user enters a negative par value.